

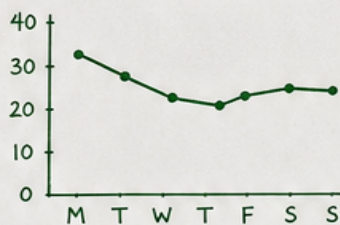
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Claim

Our community garden should water early in the morning for 20 minutes, three days each week.

Evidence

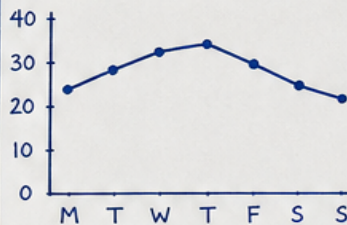
① Soil Moisture (%)
7-day average



- Morning (6-8am): 28%
- Afternoon (2-4pm): 14%
- Evening (7-9pm): 20%

Sensors: SM1, SM2, SM3
Dates: Sep 6-Sep 12

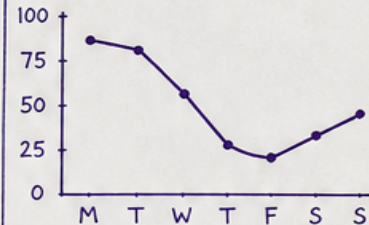
② Temperature (°C)
7-day average



- Afternoon peak: 32°C
- Morning average: 18°C
- Evening average: 22°C

Sensors: T1, T2
Dates: Sep 6-Sep 12

③ Plant Leaf Wetness (%)
7-day average



- Morning (6-8am): 85%
- Afternoon (2-4pm): 20%
- Evening (7-9pm): 45%

Sensor: LW1
Dates: Sep 6-Sep 12

Reasoning

① Soil moisture is highest in the morning and lowest in the afternoon. Watering in the morning reduces water loss to evaporation and helps plants absorb more.

② Afternoon temperatures are the highest. Watering then would increase evaporation. Morning temperatures are cooler, so water stays in the soil longer.

③ Leaf wetness is highest in the morning and lowest in the afternoon. Watering in the morning supports healthy leaves without leaving them wet overnight.

Integration

All three data sets show that early morning watering gives plants the most benefit and reduces water loss. Midday watering leads to more evaporation and less available water for plants.

Conclusion

Based on our sensor data, our community garden should water early in the morning for 20 minutes, three days each week.

Plan/
Next Steps

- Set irrigation timer for 6:30am, 20 minutes, on Mon, Wed, Fri.
- Continue collecting sensor data for the next 4 weeks.
- Re-evaluate and adjust watering time if needed.

